

Another technique to deal with priority inversion is known as priority ceiling. Priority ceiling associates a priority with each resource; the scheduler then transfers that priority to any task that accesses the resource. The priority assigned to the resource is the priority of its highest-priority user, plus one. Once a task finishes with the resource, its priority returns to normal. Both priority inversion and priority ceiling require operating system support. Linux supports priority ceiling via PTHREAD_PRIO_PROTECT protocol and priority inheritance via PTHREAD_PRIO_INHERIT protocol. For more details on techniques to avoid priority inversion, read the following paper:

<http://www-md.e-technik.uni-rostock.de/ma/gol/rtsys/articulos/YCS159.pdf>

Takeaway problem for this month

This month's takeaway problem comes from Nilesh Govande. Thank you Nilesh, for sending in the question. Given a two-dimensional NXN array of positive and negative integers, find the sub-rectangle with the largest sum. The sum of a rectangle is the sum of all the elements in that rectangle. In this problem, the sub-rectangle with the largest sum is referred to as the *maximal sub-rectangle*. A sub-rectangle is any contiguous sub-array of size 1X1 or greater, located within the whole array. As an example, the maximal

sub-rectangle of the array:

0	-2	-7	0
9	2	-6	2
-4	1	-4	1
-1	8	0	-2

...is in the lower-left-hand corner:

9	2
-4	1
-1	8

...and has the sum of 15.

If you have any favourite programming puzzles that you would like to discuss on this forum, please send them to me. Feel free to send your solutions and feedback to sandyasm_AT_yahoo_DOT_com. Till we meet again next month, happy programming! 

About the author:

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